



Module 1

INTRODUCTION COMPUTER ORGANIZATION/ ARCHITECTURE

I. INTRODUCTION

This module introduces the basic terminologies in computer organization/architecture. Computer core structure and function promotes understanding how computer work within its electronic components inside. Logic gates is a critical concept to master in able to understand computers deeply.

II. OBJECTIVES

1. Distinguish the terminologies in computer architecture/ organization
2. Analyze the core functions and structure of a computer.
3. Analyze and evaluate different logic gates.

III. LESSON PROPER

COMPUTER ORGANIZATION OVERVIEW

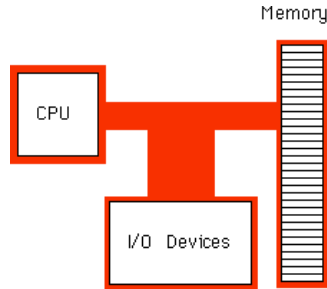
The Basic System Components

The basic operational design of a computer system is called its *architecture*. John Von Neumann, a pioneer in computer design, is given credit for the architecture of most computers in use today.

For example, the 80x86 family uses the *Von Neumann architecture* (VNA). A typical Von Neumann system has three major components:

- *central processing unit* (or *CPU*),
- *memory*
- *Input/output* (or *I/O*).

The way a system designer combines these components impacts system performance:



Computer System

- complete configuration of a computer including the peripheral units and the system programming which make it a useful and functional machine for a determined task

Central Processor (CPU)

- This part is also known as central processing unit or CPU, which in turn is made by the control unit and the arithmetic and logic unit.
- Its functions consist in reading and writing the contents of the memory cells, to forward data between memory cells and special registers, and decode and execute the instructions of a program.

Central Memory

- It is a group of cells, now being fabricated with semi-conductors, used for general processes, such as the execution of programs and the storage of information for the operations.
- The generic name of these memories is Random Access Memory or RAM.
- The main disadvantage of this type of memory is that the integrated circuits lose the information they have stored when the electricity flow is interrupted.

Input and Output Units (I/O)

- In order for a computer to be useful to us it is necessary that the processor communicates with the exterior through interfaces which allow the input and output of information from the processor and the memory.
- Through the use of these communications it is possible to introduce information to be processed and to later visualize the processed data.

Auxiliary Memory Units.

- The stored information on these magnetic media means receive the name of files.
- A file is made of a variable number of registers, generally of a fixed size; the registers may contain information or programs.

Generation of Computers

FIRST GENERATION

The 1st generation of computer in 1940's were computers whose architecture was based on massive electronic valve (vacuum tube). e.g. electronically digital vacuum computers, universal vacuum computer UNIVAC

Features:

- Use of vacuum tubes
- Big & Clumsy
- High Electricity Consumption
- Programming in Mechanical Language
- Larger AC were needed
- Lot of electricity failure occurred

SECOND GENERATION

2nd Generation computers were more reliable and faster in performance (read and write operations) than the 1st generation which were massive, instead of valves simple transistors were used in design e.g. ATLAS computers

Features:

- Transistors were used
- Core Memory was developed
- Faster than First Generation computers
- First Operating System was developed
- Programming was in Machine Language & Assembly Language
- Magnetic tapes & discs were used
- Computers became smaller in size than the First Generation computers
- Computers consumed less heat & consumed less electricity

THIRD GENERATION

Were more powerful, reliable and more compatible in which simple integrated circuits were introduced. They emerged in early 1960s, and examples are IBM 360 Series.

Features:

- Integrated circuits developed
- Power consumption was low
- SSI & MSI Technology was used
- High level languages were used

FOURTH GENERATION

4th generation computers uses sophisticated micro-electronic devices.

Features:

- LSI & VLSI Technology used
- Development of Portable Computers
- RAID Technology of data storage
- Used in virtual reality, multimedia, simulation
- Computers started in use for Data Communication

- Different types of memories with very high accessing speed & storage capacity

FIFTH GENERATION COMPUTERS

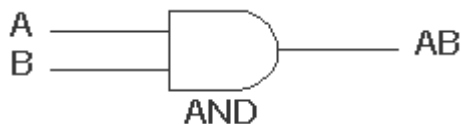
- Used in parallel processing
- Used superconductors
- Used in speech recognition
- Used in intelligent robots
- Used in artificial intelligence

Review on Logic Gates

Logic gates

- Digital systems are said to be constructed by using logic gates.
- These gates are the AND, OR, NOT, NAND, NOR, EXOR and EXNOR gates.
- The basic operations are described below with the aid of truth tables.

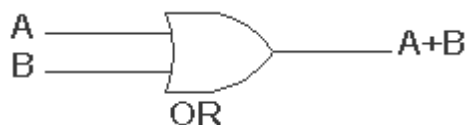
AND gate



2 Input AND gate		
A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

- The AND gate is an electronic circuit that gives a high output (1) only if all its inputs are high.
- A dot (.) is used to show the AND operation i.e. A.B.

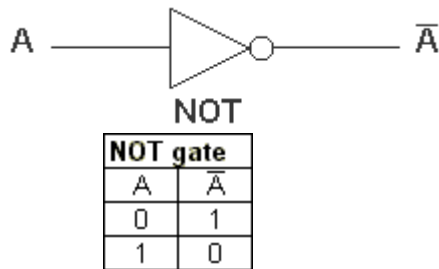
OR gate



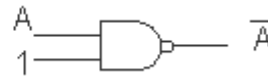
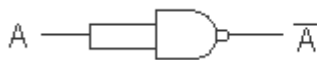
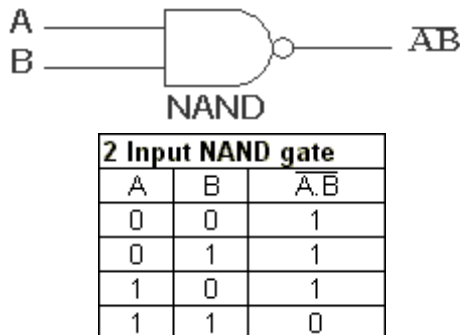
2 Input OR gate		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

- The OR gate is an electronic circuit that gives a high output (1) if **one or more** of its inputs are high.

- A plus (+) is used to show the OR operation.

NOT gate

- The NOT gate is an electronic circuit that produces an inverted version of the input at its output.
- It is also known as an *inverter*. If the input variable is A, the inverted output is known as NOT A.
- This is also shown as A', or A with a bar over the top, as shown at the outputs.

**NAND gate**

This is a NOT-AND gate which is equal to an AND gate followed by a NOT gate. The outputs of all NAND gates are high if **any** of the inputs are low. The symbol is an AND gate with a small circle on the output. The small circle represents inversion.

NOR gate



2 Input NOR gate		
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

- This is a NOT-OR gate which is equal to an OR gate followed by a NOT gate.
- The outputs of all NOR gates are low if **any** of the inputs are high.
- The symbol is an OR gate with a small circle on the output. The small circle represents inversion.

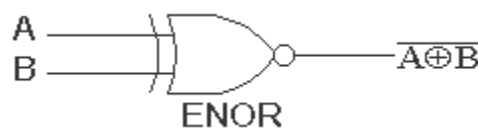
EXOR gate



2 Input EXOR gate		
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

- The '**Exclusive-OR**' gate is a circuit which will give a high output if **either, but not both**, of its two inputs are high.
- An encircled plus sign ($\oplus$) is used to show the EOR operation.

EXNOR gate



2 Input EXNOR gate		
A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

- The '**Exclusive-NOR**' gate circuit does the opposite to the EOR gate.
- It will give a low output if **either, but not both**, of its two inputs are high.
- The symbol is an EXOR gate with a small circle on the output. The small circle represents inversion.

- The NAND and NOR gates are called *universal functions* since with either one the AND and OR functions and NOT can be generated.

Note:

A function in *sum of products* form can be implemented using NAND gates by replacing all AND and OR gates by NAND gates.

A function in *product of sums* form can be implemented using NOR gates by replacing all AND and OR gates by NOR gates.

Table 1: Logic gate symbols

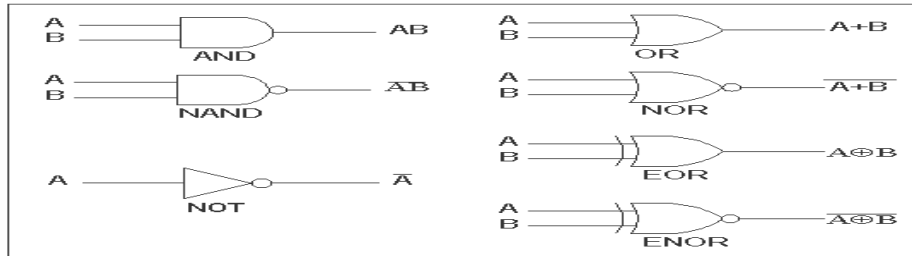


Table 2: Logic gates representation using the Truth table

NOT gate	
A	$\overline{A}$
0	1
1	0

INPUTS		OUTPUTS					
A	B	AND	NAND	OR	NOR	EXOR	EXNOR
0	0	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	0	1	0	0	1

IV. PRACTICE EXERCISES/ACTIVITIES

Logic gates: Complete the logic table

A	B	C	A'	B'	$A+B'C'$	$A'(B'C')'$	$AB+1$	$(A''' + C'')'0$	$((AB)(AC)(BC))'$
1	1	1							
1	1	0							
0	0	1							
0	0	0							
1	0	1							

V. ADDITIONAL RESOURCES

- Essential of Computer Organization and Architecture by Linda Null and Julia Lobur
- Computer Systems: Digital Design, Fundamentals of Computer Architecture and Assembly Language by Eta Elahi

VI. ASSESSMENT

To be given in a separate worksheet.

VII. REFERENCES

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